

Estimating a Time Allocation Model using Revealed and Stated Preference Data*

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Abstract

Structural models of time allocation are typically estimated using detailed time-use diaries. Because these data are expensive to collect and often unavailable, we develop an alternative identification and estimation strategy based primarily on qualitative survey responses. Specifically, we combine ordinal reports of time use, stated valuations of different activities, and a small number of quantitative anchors. We show that these data are sufficient to identify preferences over activities, the underlying production technology, and the opportunity cost of time within a penalized maximum likelihood framework. Monte Carlo simulations calibrated to our application demonstrate that the proposed estimators recover the model's key parameters accurately and perform nearly as well as estimators based on traditional time-use diaries. We apply the method to news consumption and document substantial information gaps by race, ethnicity, and education. Minority and less-educated individuals exhibit stronger preferences for local news, whereas others place greater value on national and international coverage. These disparities are driven primarily by differences in wages and preferences rather than by unequal access to news providers.

Keywords: Time allocation, information acquisition, qualitative survey data, ordered response models, nonlinear estimation, opportunity cost of time, news consumption, information inequality.

JEL Codes: C25, C83, D83, J22.

1 Introduction

In empirical economics, identification and estimation of behavioral models have traditionally relied on revealed preference data.¹ These approaches are typically based on observed choices and objectively measured variables such as prices and individual characteristics, and have proven highly successful across a wide range of applications. However, when key determinants of behavior, such as beliefs, information, and preferences, are only imperfectly observed, revealed preference methods face inherent limitations. This has led to a new literature that incorporates alternative data sources into the empirical analysis, among them subjective beliefs, stated preferences, and other qualitative survey responses. Such data complement traditional revealed-preference sources ([Almås et al., 2024](#)).

We show that a structural model of time allocation with endogenous information acquisition can be estimated without detailed time diaries, using qualitative survey responses instead. We primarily rely on ordinal categorical reports of time use data combined with stated valuations of news acquisition activities. Building on [Becker \(1965\)](#) and subsequent work in labor economics, individuals allocate a finite time endowment across market and non-market activities, including the acquisition of information from multiple sources. Following [Gronau \(1977\)](#) and [Aguiar and Hurst \(2016\)](#), we adopt a two-stage framework in which wages determine total non-market time, while preferences govern its allocation across activities. Behavior in this framework reflects three distinct structural objects: preferences over news types, which vary with demographics; news-production technologies that transform time inputs into information; and the opportunity cost of time.

The central econometric challenge is to identify these objects separately, because observed time allocations confound them. An individual may spend little time on news because she values it little, because her sources deliver information efficiently, or because her time is expensive. Welfare statements and counterfactual analysis require the separation: attributing group differences in news consumption to preferences, to technology, or to the price of time leads to very different conclusions about

¹Seminal contributions were made by [Samuelson \(1938, 1948\)](#) and [Arrow \(1959\)](#).

the sources of informational inequality. The same challenge arises whenever time is an input into a valued non-market output, as in home production, parental time investment, or health, so the econometric approach developed here applies beyond the news application.

Whether these objects can be separated, and how precisely, depends on the data available. Because our estimators combine data sources in nonstandard ways, we evaluate them in a Monte Carlo experiment matched to our application, asking what each data configuration can recover and with what precision. The experiment compares four estimators that are ordered by the demands they place on time-use measurement. The first observes complete time diaries, the richest revealed-preference data one could hope to collect; the second observes only marginal totals by provider and by news type, the format in which time-use data are typically available in most time diaries such as the American Time Use Survey.

The remaining estimators correspond to the setting of our application, in which time diaries are not available. The third estimator replaces the diaries with ordinal survey responses that record how often each provider is used and how closely each news type is followed, together with a small set of aggregate quantitative anchors. The ordinal responses are revealed-preference data but carry no cardinal scale; the anchors supply that scale.² The fourth adds stated-preference responses that rate how much respondents value different news topics; these responses do not describe behavior but measure the intensity of preferences directly. Comparing these identification and estimation strategies helps us to assess advantages of the different data structures.

The Monte Carlo experiment delivers two conclusions. First, the time-use data identify the key parameters of the model at either level of aggregation; the cost of aggregation is a loss of precision, not of identification. Second, the survey-based estimators recover the same objects: ordinal responses combined with a few aggregate anchors substitute for time diaries, and adding a parsimonious stated-preference index sharpens the estimates further, roughly halving the sampling error of the survey-only estimator and narrowing the gap to the precision that continuous time diaries would

²Although qualitative responses lack the numerical precision of quantitative measures, they are typically easier for respondents to provide, allowing researchers to collect detailed information on a broader range of activities (Stantcheva, 2023). They are also considerably less costly to collect.

provide. We, therefore, conclude that our novel estimators that are primarily based on categorical survey and stated preference data perform sufficiently well compared to estimators that exploit “ideal” time diary data. However, our data are much easier and cheaper to collect than traditional time use data and are feasible when traditional data do not exist.

We bring the estimators to two complementary data sources. The primary source, the Local News Survey (LNS), provides qualitative categorical measures: ordinal reports of how closely individuals follow local, national, and international news, how often they use each news provider, and how important they find different local news topics. The second, the Media Consumption Survey (MCS), provides traditional quantitative time-use data, namely the average minutes spent on news each day, by age. The categorical responses in the LNS identify the model’s parameters, while the aggregate time-use anchors from the MCS fix their scale.

Our empirical findings reveal significant differences in news consumption across race, ethnicity, and education. African Americans spend the most time on local news (74 minutes per day), followed by Hispanics (47 minutes) and whites (24 minutes). These differences are economically meaningful when valued by wages. In contrast, whites devote the most time to national news, while international news receives the least attention, though Hispanics allocate slightly more time than others. Decomposing these gaps shows that differences in wages and preferences, rather than access to news providers, primarily drive these patterns. Minorities tend to have lower opportunity costs of time and stronger preferences for local news, whereas whites favor national and international news, partially offset by higher time costs. These trends also extend to differences by education, highlighting that news consumption is shaped more by personal economic factors and preferences than by access to content. These demographic patterns in local news interest are consistent with descriptive evidence from the political communication literature.³ The model, therefore, enables welfare-relevant comparisons and counterfactual policy analysis, which improve our understanding of inequality in news acquisition.

In summary, this paper illustrates how qualitative survey data can identify and

³See, for example, [Prior \(2007\)](#) and [Pew Research Center \(2019\)](#).

estimate structural models in environments where the key variables are imperfectly measured. By combining low-cost categorical measures with a small number of quantitative anchors, we develop a feasible estimation strategy of a model of time allocation and news acquisition. Our empirical findings provide new insights into differences in news consumption across race, ethnicity, and education.

Our work contributes to several strands of the literature on applied econometrics and labor economics. First, the paper is related to a literature that uses survey-based measures to identify and estimate structural models. [Manski \(2004\)](#) argues for eliciting subjective and stated responses and using them as measurements in estimation. Many papers now use elicited beliefs, expectations, and stated preferences to identify preference and choice parameters, including [Dominitz and Manski \(1997\)](#) on income expectations, [Cunha et al. \(2013\)](#) on maternal expectations, [Wiswall and Zafar \(2018\)](#) on workplace preferences, [Maestas et al. \(2023\)](#) on the value of job attributes, [Blass et al. \(2010\)](#) on random-utility models from elicited choice probabilities, and [van der Klaauw \(2012\)](#) and [Arcidiacono et al. \(2012\)](#) on expectations and elicited counterfactuals in structural choice models. [Almås et al. \(2024\)](#) make a broader case for new survey-based measures in models of decision-making. We add to this work by showing that coarse categorical responses, not just numerical expectations, can identify and estimate a structural time-allocation model.

Our work also builds on two areas of applied econometrics. The first is the literature on combining data sources. [Imbens and Lancaster \(1994\)](#) and [Ridder and Moffitt \(2007\)](#) show how moments from one dataset can supplement another, and two-sample and micro-plus-macro estimators identify parameters that no single dataset pins down ([Angrist and Krueger, 1992](#); [Berry et al., 2004](#); [Chen et al., 2008](#); [Hwang, 2026](#)). We use this idea in our setting: the categorical responses identify the model’s parameters, and a few aggregate anchors fix their scale. The second is the measurement of latent variables from ordered categorical responses. We treat the survey answers as noisy measures of latent time use and preferences, using the ordered-response models of [Andrich \(1978\)](#), [McCullagh \(1980\)](#), and [Greene and Hensher \(2010\)](#). We are closest to work that treats categorical survey responses as noisy measures of latent variables ([Cunha et al., 2010](#); [Munro and Ng, 2022](#); [Oparina and Srisuma, 2022](#)).

Second, the paper is related to research in labor economics on time use. The theory goes back to [Becker \(1965\)](#) and [Gronau \(1977\)](#), and [Ghez and Becker \(1975\)](#) and [Juster and Stafford, eds \(1985\)](#) are early empirical work. A large structural literature then estimates preferences and technologies from time-use data, including [Kooreman and Kapteyn \(1987\)](#) and [Biddle and Hamermesh \(1990\)](#) on time allocation, [Aguilar and Hurst \(2007\)](#) on trends in leisure and work, [Fiorini and Keane \(2014\)](#) on children’s time and skills, [Blundell et al. \(2018\)](#) on insurance against wage shocks, [Rogerson and Wallenius \(2019\)](#) on older couples, [Bastian and Lochner \(2022\)](#) on responses to the earned income tax credit, and, in family economics, [Chiappori et al. \(2002\)](#), [Cherchye et al. \(2012\)](#), and [Lise and Yamada \(2019\)](#) on household preferences and bargaining. This work relies on detailed time diaries, and [Kerkhofs and Kooreman \(2003\)](#) and [Chou and Shi \(2021\)](#) discuss what such models can identify from the data at hand. We contribute by estimating a structural time-use model without diaries, from qualitative survey responses instead. This matters because diaries are costly to collect and, for news, unavailable: the American Time Use Survey does not record time spent on news, while rich tracking data exist for online news consumption (e.g., [Gentzkow and Shapiro, 2011](#); [Levy, 2021](#)).

The rest of the paper is organized as follows. Section [2](#) introduces our data sets and discusses the survey design as well as the main survey questions used in the analysis. We also summarize the reduced form evidence regarding the observed gaps in news acquisition by race, ethnicity, and education. Section [3](#) develops our time-use and information acquisition model. Section [4](#) discusses the identification and estimation of the parameters of the model. Section [5](#) reports our empirical results and provides a detailed analysis of the informational gaps. Section [6](#) offers some conclusions and discusses future research.

2 Data

2.1 The Local News Survey

Our primary dataset is the Local News Survey (LNS), a large, nationally representative survey conducted by the Pew Research Center between October 15 and November 8, 2018. It draws on two probability-based online panels of U.S. adults, the American Trends Panel and Ipsos KnowledgePanel, and covers non-institutionalized adults across 932 core-based statistical areas, with responses collected by web, mail, and telephone. Of the 34,897 respondents (a 56% response rate), 27,563 provide complete demographic and socioeconomic information, and survey weights make the sample nationally representative. For each respondent, the LNS records detailed demographics (age, gender, race and ethnicity, education, marital status, political affiliation, and income) together with measures of local community attachment and perceived neighborhood quality.⁴

Table 1 summarizes the sample. It is close to evenly split by gender and spans all age groups, and, most relevant for our analysis, it contains sizable racial and ethnic minorities alongside a wide spread of education: 64 percent White, 16 percent Hispanic, and 12 percent African American, with about a third college graduates. Reported income spans the full range, and the average imputed hourly wage is \$19.94.

The estimation of our time allocation model is primarily based on three qualitative survey questions that are central to this survey. First, the LNS elicits qualitative answers on time allocations on different types of news using the following question: *How closely do you follow ...?* for international, national, and local news. Responses are recorded on a four-point Likert scale (*not at all closely, not very closely, somewhat closely, very closely*) and are reported retrospectively for three-week periods in 2018, 2017, and 2016. Unlike standard time-use surveys, the LNS relies on ordered categorical measures rather than quantitative time allocations. This reduces reporting burden and may mitigate recall error, but it does not provide cardinal measures of

⁴Note that income is reported in brackets. We also estimate a wage equation using the Current Population Survey and impute log hourly wages based on observable characteristics. The resulting mean hourly wage is 19.9 (s.d. 4.65), consistent with the income distribution in the data.

Table 1: Descriptive Statistics of LNS

<i>Age</i>		<i>Marital Status</i>	
18-29	0.209	Married	0.483
30-49	0.348	<i>Party Affiliation</i>	
50-64	0.262	Republican	0.263
65+	0.181	Democrat	0.331
<i>Gender</i>		Independent	0.276
Male	0.490	Other	0.130
Female	0.510	<i>Income</i>	
<i>Education</i>		Less than \$10,000	0.097
College Graduate	0.314	\$10,000 to less than \$20,000	0.100
Some College Education	0.321	\$20,000 to less than \$30,000	0.115
High School Graduate	0.276	\$30,000 to less than \$40,000	0.104
<i>Race</i>		\$40,000 to less than \$50,000	0.102
White	0.644	\$50,000 to less than \$75,000	0.166
African American	0.116	\$75,000 to less than \$100,000	0.124
Hispanic	0.159	\$100,000 to less than \$150,000	0.116
Others	0.081	\$150,000 or more	0.076
<i>Local Community Attachment</i>		<i>Local Community Rating</i>	
Very much	0.225	Excellent	0.312
Somewhat	0.485	Good	0.550
Not very	0.228	Only fair	0.118
Not at all	0.061	Poor	0.020
<i>Hourly Wages</i>			
Mean	19.94		
St. Dev.	4.65		

Notes: Survey-weighted shares of the LNS analysis sample ($N = 27,563$), Pew Research Center Local News Survey. Hourly wages are imputed from the CPS as described in the text.

time use.

Table 2: Exposure to Local, National, and International News

	<i>How Closely Do You Follow?</i>		
	Local	National	International
African American	0.863*** (0.048)	−0.038 (0.047)	−0.082* (0.046)
Hispanic	0.428*** (0.041)	0.215*** (0.042)	0.432*** (0.041)
College Grad	−0.604*** (0.076)	0.712*** (0.074)	0.457*** (0.074)
Age	Yes	Yes	Yes
Income	Yes	Yes	Yes
Political Affiliation	Yes	Yes	Yes
Gender and Marital Status	Yes	Yes	Yes
City FE	Yes	Yes	Yes

Notes: Ordered logit coefficients from the LNS; the outcome is the reported closeness of following each news type. Standard errors in parentheses. Controls are included as indicated; all specifications include city fixed effects. *, **, and *** denote significance at the 10, 5, and 1 percent levels.

One of the key empirical objectives of this paper is to characterize differences in news acquisition by race, ethnicity, and education. Hence, we start by estimating ordered logit models of attention to local, national, and international news, controlling for demographics, political affiliation, community characteristics, and city fixed effects. The results are summarized in Table 2. The results reveal systematic differences across socio-demographic groups. African Americans and less-educated individuals are significantly more likely to follow local news, whereas more-educated individuals devote relatively more attention to national and international news. The magnitude is economically meaningful: the coefficient for African Americans in local news implies an odds ratio above two relative to whites. Hispanics exhibit higher engagement across all categories. These patterns are stable, and in some cases strengthen, when additional controls are included.

Second, the LNS also characterizes the sources from which individuals obtain local news. It distinguishes five provider types: printed newspapers, television, radio,

social media (e.g., Facebook, YouTube, Snapchat), and online media. Importantly, this classifies news by the platform used to access it, not by who originally produced the content. Respondents report how often they use each source (*often*, *sometimes*, *hardly ever*, *never*), again yielding ordered categorical measures.

Table 3: Time Allocations Among Local News Providers

	Newspaper	Radio	TV	Online	Social Media
African American	0.11** (0.05)	0.10** (0.04)	0.73*** (0.05)	0.06 (0.05)	0.09* (0.04)
Hispanic	-0.03 (0.04)	-0.09** (0.04)	0.23*** (0.04)	0.10** (0.04)	0.27*** (0.04)
College Grad	0.12 (0.07)	0.34*** (0.07)	-0.59*** (0.08)	0.61*** (0.07)	-0.41*** (0.07)
Age	Yes	Yes	Yes	Yes	Yes
Income	Yes	Yes	Yes	Yes	Yes
Political Affiliation	Yes	Yes	Yes	Yes	Yes
Gender and Marital Status	Yes	Yes	Yes	Yes	Yes
Community Characteristics	Yes	Yes	Yes	Yes	Yes
City FE	Yes	Yes	Yes	Yes	Yes

Notes: Ordered logit coefficients from the LNS; the outcome is the reported frequency of use of each local news provider. Standard errors in parentheses. Controls are included as indicated; all specifications include city fixed effects. *, **, and *** denote significance at the 10, 5, and 1 percent levels.

We estimate ordered logit models of time allocation across providers (Table 3), controlling for demographics, political affiliation, community characteristics, and city fixed effects. African Americans and Hispanics allocate more time to local news providers than whites do. African Americans exhibit significantly higher usage across most platforms, with particularly large effects for television (implying an odds ratio of approximately 2.08), as well as positive and significant differences for newspapers, radio, and social media. Hispanics also show higher usage, especially for television and social media, though the magnitudes are smaller. Differences by education are more nuanced: more-educated individuals allocate more time to online media and, to a lesser extent, traditional sources such as radio and newspapers, while less-educated individuals rely more on television and social media.

Table 4: Preferences for Local News Topics

	Crime	Politics	Community
African American	0.87*** (0.05)	0.32*** (0.05)	0.35*** (0.05)
Hispanic	0.62*** (0.04)	0.28*** (0.04)	0.20*** (0.04)
College Grad	-0.72*** (0.08)	0.50*** (0.08)	0.24*** (0.08)
	Jobs	Schools	Economy
African American	1.00*** (0.05)	0.93*** (0.05)	0.87*** (0.05)
Hispanic	0.77*** (0.04)	0.64*** (0.04)	0.69*** (0.04)
College Grad	-0.35*** (0.08)	-0.38*** (0.07)	-0.49*** (0.08)
	Sports	Culture	Restaurants
African American	0.68*** (0.04)	0.05 (0.05)	0.20*** (0.05)
Hispanic	0.33*** (0.04)	0.28*** (0.04)	0.13*** (0.04)
College Grad	-0.17** (0.07)	0.68*** (0.08)	0.21*** (0.07)
	Traffic	Weather	Importance index
African American	0.57*** (0.05)	0.12* (0.06)	0.98*** (0.04)
Hispanic	0.44*** (0.04)	-0.02 (0.05)	0.71*** (0.04)
College Grad	-0.06 (0.08)	0.09 (0.09)	-0.07 (0.07)
Age	Yes	Yes	Yes
Income	Yes	Yes	Yes
Political Affiliation	Yes	Yes	Yes
Gender and Marital Status	Yes	Yes	Yes
Community Characteristics	Yes	Yes	Yes
City FE	Yes	Yes	Yes

Notes: Ordered logit estimates; standard errors in parentheses. The final column uses the importance index, the average of the eleven topic responses, as the outcome. *, **, and *** denote significance at the 10, 5, and 1 percent levels.

Finally, the LNS elicits preferences over local news topics. The survey covers eleven topics, including politics, crime, education, jobs, the local economy, culture, sports, weather, and traffic. Importance is measured on an ordered scale ranging from low salience to daily relevance. We analyze heterogeneity in topic preferences in Table 4. Racial and ethnic differences are pronounced across economically salient categories such as crime, jobs, schools, and the local economy, while differences are smaller for culture and weather. Education differences follow a similar pattern: less-educated individuals place greater importance on economically relevant topics, whereas more-educated individuals emphasize politics and cultural content. These patterns are consistent with the differences in news consumption documented above.

For the structural analysis below we summarize the eleven topic responses by their average, after coding each response so that higher values indicate greater importance: $\bar{K}_i = \frac{1}{11} \sum_{m=1}^{11} k_{im}$, with $k_{im} \in \{1, \dots, 4\}$. The average ranges from 1 to 4 (mean 2.8, standard deviation 0.5) and, being built from eleven items, is far less noisy than any single response; the eleven items form a coherent scale (Cronbach's $\alpha = 0.79$). The structural measurement model below discretizes this average into four ordered categories of local news importance, from low to high salience. Item nonresponse is rare (about one percent per topic); missing items are imputed by ordered logit on demographics, following the same convention used in the construction of the grouped aggregates.

Because a one-dimensional summary could in principle average away the group differences documented in Table 4, the final column of the table reports the same ordered logit specification with the index as the outcome. The index preserves the heterogeneity: the coefficients for African Americans (0.98) and Hispanics (0.71) are comparable to the largest topic specific gradients, those for jobs, schools, and the local economy. The college gradient is small and statistically insignificant for the index, because the positive gradients for politics and cultural topics offset the negative gradients for the economically salient topics. The single index therefore retains the racial and ethnic heterogeneity in stated preferences that the structural model exploits.

We have presented reduced-form evidence on heterogeneity in news acquisition

across demographic groups. Overall, the evidence indicates substantial and robust disparities in news acquisition across demographic groups. However, the qualitative nature of the survey measures limits our ability to interpret these differences. In particular, the data do not provide cardinal measures of time or resources allocated to news consumption. Addressing this limitation requires additional data and an econometric approach that combines these different data sources.

2.2 The Media Consumption Survey

Our second data source is the Media Consumption Survey (MCS), a biennial survey of 3,003 U.S. adults. We focus on the 2012 wave, which is the most recent one Pew fielded with this information. Although it predates the LNS by several years, time spent on news is fairly stable across the earlier waves (Table 5), so we take the 2012 levels as representative of our sample period. The weighted demographic composition of the MCS closely matches that of the LNS, with minor differences in educational attainment.

Table 5: Average Time-use in Minutes in the MCS

	2004	2006	2008	2010	2012	Average
Total	72	69	66	70	67	69
Age 18-29	45	49	46	45	45	46
Age 30-39	70	65	63	68	62	66
Age 40-49	73	64	67	74	71	70
Age 50-64	82	76	74	81	76	78
Age 65+	88	79	84	83	83	83

Notes: Average minutes per day spent on news by age group and survey wave, Pew Research Center News Consumption Survey. The 2012 wave provides the age-group anchors used in the estimation.

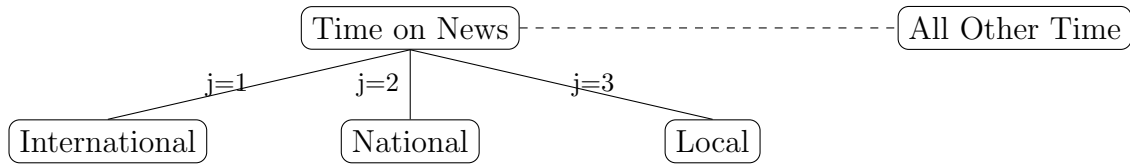
The MCS provides information on the daily time allocated to news consumption. On average, individuals spend 66–72 minutes per day on news, with substantial heterogeneity by age (45 minutes for ages 18–29 versus 83 minutes for ages 65+). These patterns are stable over time (2004–2012). We use these data to identify and estimate the scale of time allocations in the model.

3 A Time Allocation and News Acquisition Model

The data tell us how much time people spend on each news provider and how important they say local news is, but they do not directly reveal the preferences or the opportunity cost of time behind these choices. To recover these, we build a structural model of how an individual divides a fixed time budget across the three news types, local, national, and international, and across the providers within each type, trading news consumption against all other uses of time. Estimating the model recovers two objects of interest: the preference weights on each news type and the opportunity cost of time. With these in hand, we decompose group differences in news consumption into a part driven by preferences and a part driven by wages, and we compute willingness to pay for news. The model is also the basis for the paper’s main question, how much each type of data reveals about the opportunity cost of time, which we take up in Section 4.

We consider a model in which the information structure can be partitioned into a simple nesting structure. The first layer of the structure consists of the different types of news. In our application, there are three types: local, national, and international news. Let J denote the number of news types. The information structure is illustrated in Figure 1.

Figure 1: Information Structure



There are a number of different news providers that produce content. Individuals allocate their time among these providers. Let $\mathcal{S}$ denote the set of news providers. In our empirical application, there are five types of providers: radio, television, printed newspaper, online media, and social media. Let $|\mathcal{S}|$ be the maximum number of providers available that could be in a consumer’s choice set.⁵ We indicate by $S \subset \mathcal{S}$ the set of news providers that are available to an individual. Let $|S|$ denote the

⁵In our application $|\mathcal{S}| = 5$.

number of providers in bundle S . For example, an individual may have access to radio, television, social media, and online media, but does not subscribe to a printed newspaper, hence $S = \{radio, television, online\ media, social\ media\}$ and $|S| = 4$. We take these bundle choices as given and study the time allocation among the providers for each type.⁶

We solve the optimal time allocation and information acquisition problem sequentially. First, we derive the optimal time allocations among providers for an arbitrary time allocation among news types. Second, we characterize the optimal time allocation among news types and derive the optimal time allocated to total news acquisition.

Consider an individual with a predetermined time budget H_j that has been allocated to news type j . Let h_{js} denote the time that the individual spends on service provider s . Hence, $h_j = \{h_{js}\}_{s \in S}$ denotes the full time allocation vector for news type j . The time allocation choices of an individual must satisfy the following constraints:

$$\begin{aligned} \sum_{s \in S} h_{js} &\leq H_j \\ h_{js} &\geq 0 \text{ if } s \in S \\ h_{js} &= 0 \text{ if } s \notin S \end{aligned} \tag{1}$$

A time allocation vector translates into a vector of news or information acquisition. The total news production for type j is denoted by $t_j(S, h_j)$ and depends on the bundle choice and the time allocation vector. We assume that:

$$t_j(S, h_j) = \sum_{s \in S} t_{js} f_j(h_{js}) \tag{2}$$

where $f_j(\cdot)$ is strictly concave, differentiable, and strictly monotonically increasing in h_{js} . Moreover $f_j(0) = 0$. Note that the parameters t_{js} capture the relative advantages of news providers in certain types.⁷ News production is additively separable across providers.⁸ The concavity in the news production generates an interior solution for

⁶We discuss in the conclusions how to extend our model to account for endogenous bundle choices.

⁷This specification also imposes the normalizing assumption that $t_j(\cdot, \emptyset) = 0$.

⁸Crawford and Yurukoglu (2012) also impose this assumption. However, this separability as-

the time allocation problem. For our empirical model, we assume that

$$f_j(h_{js}) = \frac{1}{1-\rho} h_{js}^{1-\rho} \quad (3)$$

where $\rho > 0$ governs the curvature of news production.⁹

Let x denote an observed vector of individual characteristics that shift preferences. Let $U_j(S, x, h_j)$ denote the utility of news type j associated with a bundle S and time allocation vector h_j for an individual with characteristics x . The utility of news type j is proportional to its news production $t_j(S, h_j)$, scaled by a preference weight $\gamma_j(x)$:

$$U_j(S, x, h_j) = \gamma_j(x) \sum_{s \in S} t_{js} f_j(h_{js}) \quad (4)$$

where $\gamma_j(x)$ captures heterogeneity in preferences for type j or the intensity with which individuals consume type j . For example, some individuals pay more attention to national than international news. In the empirical model, we assume that $\gamma_j(x) = \exp(x' \gamma_j)$.¹⁰

Given a pre-determined time budget H_j , individuals optimally allocate the time among the providers in their choice sets. Hence, individuals maximize utility in equation (4) subject to the time constraints in equations (1). The Lagrangian for this optimization problem can be written as:

$$\max \quad \gamma_j(x) \sum_{s \in S} t_{js} f_j(h_{js}) + \mu_j \left(H_j - \sum_{s \in S} h_{js} \right) \quad (5)$$

where μ_j is the Lagrange multiplier for news type j . For $s \in S$, the first-order

sumption can, in principle, be relaxed and is primarily made to obtain a tractable solution for the time allocation problem.

⁹Although the model is developed for this general specification, we impose the logarithmic case in the empirical application and in the Monte Carlo experiments for simplicity. The logarithmic production function arises as the limit of equation (3) as ρ approaches one, up to a constant that does not affect behavior. The restriction is supported by the data: when we estimate the curvature parameter in the application, the estimates suggest a value of ρ close to one.

¹⁰One could also add a random component to the preference structure here.

conditions can be written as follows:

$$\gamma_j(x) f'_j(h_{js}) t_{js} - \mu_j = 0 \quad (6)$$

while for $s \notin S$ we have $h_{js} = 0$. Solving equation (6) for h_{js} we obtain for each $s \in S$:

$$h_{js} = f_j'^{-1} \left(\frac{\mu_j}{\gamma_j(x) t_{js}} \right) \quad (7)$$

Note that we can rule out corner solutions under the assumptions we made above.¹¹ We can obtain closed-form solutions for h_{js} for a class of production functions that satisfy strict monotonicity and differentiability conditions. Consider, for example, the specification of the news production function we use in the empirical analysis in equation (3). It is straightforward to show that the solution for the optimal time allocation among providers for type j is given by:

$$h_{js}(S, x, H_j) = \frac{(\gamma_j(x) t_{js})^{\frac{1}{\rho}}}{\sum_{s' \in S} (\gamma_j(x) t_{js'})^{\frac{1}{\rho}}} H_j \quad (8)$$

Note that the time allocation is linear in H_j and the weights associated with news provider s depend on the efficiency of news production t_{js} as well as the individual preferences for news types $\gamma_j(x)$. For example, if television is good at covering local news, and the individual cares about local news, the individual allocates a higher fraction of her time to television.

The maximum utility for news type j attainable from bundle S and time endowment H_j , denoted by $U_j(S, x, H_j)$, is given by:

$$U_j(S, x, H_j) = \gamma_j(x) \sum_{s \in S} t_{js} f_j(h_{js}(S, x, H_j)) \quad (9)$$

¹¹Our model can be interpreted as a representative agent framework conditional on (x, S, w) , with idiosyncratic variation captured by the error terms in estimation. Since we only have qualitative time allocation data at the topic or provider level, we cannot investigate corner solutions at the individual level.

In our empirical specification, we obtain the following closed-form solution:

$$U_j(S, x, H_j) = u_j(S, x) \frac{1}{1-\rho} H_j^{1-\rho} \quad (10)$$

where

$$u_j(S, x) = \gamma_j(x) \sum_{s \in S} t_{js} \left(\frac{(\gamma_j(x) t_{js})^{\frac{1}{\rho}}}{\sum_{s' \in S} (\gamma_j(x) t_{js'})^{\frac{1}{\rho}}} \right)^{1-\rho} \quad (11)$$

These equations then completely characterize the optimal allocation of time among providers for each news type for an arbitrary vector of time budgets. Note that the utility of news type j is concave in H_j , which helps to obtain an interior solution to the full time allocation problem discussed below.

Next, we discuss how to allocate time among the different news types in the first layer of the information structure. Let H denote the total time endowment devoted to news consumption. Recall that $(H_1, \dots, H_J)$ describes the time allocation vector for the J news types. This vector needs to satisfy the following time budget constraint:

$$H = \sum_{j=1}^J H_j \quad (12)$$

Assuming separability among types,¹² the total utility from the time allocation vector $(H_1, \dots, H_J)$ is then given by:

$$U(S, x, H) = \sum_{j=1}^J U_j(S, x, H_j) \quad (13)$$

In our parametric model, $U_j(S, x, H_j)$ is given by equation (10). We can derive the optimal budgets allocated across news types j by solving the following decision

¹²This separability across types, together with the separability across providers assumed above, is what makes the sequential solution tractable. Appendix A relaxes this additive aggregation to a CES aggregator with its own substitution elasticity and shows that the model stays closed form and the additional parameter is separately identified.

problem:

$$\max_{H_1, \dots, H_J} \sum_{j=1}^J U_j(S, x, H_j) + \mu \left(H - \sum_{j=1}^J H_j \right) \quad (14)$$

Solving this problem, we obtain for our specification:

$$H_j(S, x, H) = \frac{u_j(S, x)^{1/\rho}}{\sum_i u_i(S, x)^{1/\rho}} H \quad (15)$$

$H_j(S, x, H)$ is thus linear in H and increasing in $u_j(S, x)$ holding the other utilities constant. Note that the optimal decision rules $H_j(S, x, H)$ depend on the full set of preferences over the two dimensional nesting structure and the effectiveness of the news providers for each type.

To derive the optimal time allocated to news consumption, we assume that each individual has a total time endowment that can be normalized to be equal to one. Time can be allocated between market time (labor supply) and non-market time H (news consumption); since the endowment is one, market time is $1 - H$, compensated at a constant wage rate of w . Preferences are defined over news consumption and a numeraire good. Let's assume that the utility function is quasi-linear in the numeraire good. Then the decision problem that characterizes the optimal allocation of time is:

$$\max_H \beta U(S, x, H) + w (1 - H) \quad (16)$$

In our parametric model, we have

$$\begin{aligned} U(S, x, H) &= \sum_{j=1}^J u_j(S, x) \frac{1}{1 - \rho} \left(\frac{u_j(S, x)^{1/\rho}}{\sum_i u_i(S, x)^{1/\rho}} H \right)^{1-\rho} \\ &= \left(\sum_{j=1}^J u_j(S, x) \left(\frac{u_j(S, x)^{1/\rho}}{\sum_i u_i(S, x)^{1/\rho}} \right)^{1-\rho} \right) \frac{1}{1 - \rho} H^{1-\rho} \\ &= u(S, x) \frac{1}{1 - \rho} H^{1-\rho} \end{aligned} \quad (17)$$

Computing the first order condition and solving this problem for our specification,

we obtain

$$H(S, x, w) = \left(\beta \frac{u(S, x)}{w} \right)^{1/\rho} \quad (18)$$

Hence H is increasing in $u(S, x)$ and decreasing in the wage w , i.e., the wage measures the opportunity cost for time spent on non-market time, such as news or information acquisition.

Substituting (18) into equation (15) gives us the optimal time allocation to news type j as a function of the wage w :

$$H_j(S, x, w) = \frac{u_j(S, x)^{1/\rho}}{\sum_i u_i(S, x)^{1/\rho}} \left(\beta \frac{u(S, x)}{w} \right)^{1/\rho} \quad (19)$$

and finally, the allocation for type j and provider s is given by:

$$h_{js}(S, x, w) = \frac{(\gamma_j(x) t_{js})^{\frac{1}{\rho}}}{\sum_{s' \in S} (\gamma_j(x) t_{js'})^{\frac{1}{\rho}}} \frac{u_j(S, x)^{1/\rho}}{\sum_i u_i(S, x)^{1/\rho}} \left(\beta \frac{u(S, x)}{w} \right)^{1/\rho} \quad (20)$$

and the marginal distribution of time allocated to providers is given by:

$$H_s(S, x, w) = \sum_j h_{js}(S, x, w) \quad (21)$$

Hence, the optimal time allocation depends on three terms. The first term characterizes the relative efficiency of provider s for type j . The second term characterizes the relative importance of news type j among all news types. And finally, the third term characterizes the importance of time allocated to news relative to market time.

In summary, we follow [Gronau \(1977\)](#) and [Aguiar and Hurst \(2016\)](#) and model time allocation using a two-stage structure. In the first stage, wages determine the total amount of time allocated to non-market activities by shaping the trade-off between market and non-market activities. In the second stage, conditional on this total non-market time, individual preferences govern how time is allocated across different non-market activities. This allows us to characterize the optimal time allocations for news for heterogeneous individuals. For our specification of the model, differences

in news consumption are driven by heterogeneity in the opportunity costs of time measured by the wage (w), preferences for news types ($\gamma_j(x)$), and access to news providers (S). Finally, note that we derived the solution to the model for one specific specification. However, we would like to point out that the estimation strategy developed below does not rely on these specific functional form assumptions and, with small modifications, can be applied to a large class of time allocation models.

4 Identification and Estimation

4.1 A GMM Estimator Based on Quantitative Time Use Data

Observe h_{js}

To fix ideas, it is useful to consider how the model could be estimated under ideal data conditions with access to precise quantitative time-use measures. Suppose that we observe time allocated to local, national, and international news at the provider level, denoted by h_{js} . Let us assume that

$$\ln h_{js} = \ln h_{js}(S, x, w \mid \theta) + \xi_{js} \quad (22)$$

where $E[\xi_{js} \mid S, x, w] = 0$. The error term ξ_{js} captures measurement error in recorded time use and idiosyncratic shocks to realized allocations after the optimization problem has been solved. Because it enters additively and is mean-independent of (S, x, w) , it represents variation orthogonal to the channels the model captures. Appendix B develops an extension in which unobserved heterogeneity in tastes enters through $\gamma_j(x)$. We can, therefore, identify and estimate the parameters of the model using GMM.

Observe H_s and H_j

Alternatively, we only observe the marginal distributions:

$$\ln H_j = \ln H_j(S, x, w \mid \theta) + \xi_j \quad (23)$$

where $\ln H_j(S, x, w \mid \theta)$ is derived from equation (19). Similarly, define

$$\ln H_s = \ln H_s(S, x, w \mid \theta) + \xi_s \quad (24)$$

where $\ln H_s(S, x, w \mid \theta)$ is derived from equation (21). Again assume that $E[\xi_j \mid S, x, w] = 0 = E[\xi_s \mid S, x, w]$. If we impose the conditional mean restrictions above, the parameters can be estimated by GMM.

The two estimators based on quantitative time use data serve as benchmarks for the survey based estimators that follow. The first corresponds to the richest revealed preference data one could collect, a complete time diary for every news type and provider combination. The second corresponds to the format in which time diaries are typically available, namely marginal totals by provider and by news type. Comparing the two isolates the cost of aggregation, and comparing both with the survey based estimators below isolates the informational content of the qualitative and stated preference data. In the Monte Carlo experiments below we implement both estimators by maximum likelihood: we assume log normal measurement errors, under which the log likelihood reduces to a nonlinear least squares criterion built on the conditional mean restrictions above, and we estimate the error variance jointly with the model parameters. The method of moments presentation in this subsection serves to make the source of identification transparent: the parameters are identified by the conditional mean restrictions alone, without the distributional assumption.

Normalization. The time allocation data cannot separately identify the level of the opportunity cost of time and the level of the composite productivities. Under the log production specification, every optimal allocation depends on these parameters only through the products βt_{js} : scaling all composites up and scaling β down by the

same factor leaves every predicted allocation and every choice probability unchanged. The stated preference responses do not resolve the indeterminacy either, because the rescaling shifts the latent index $\ln U_l$ by a constant that the location of the measurement cut-points absorbs. One normalization is therefore required. We normalize the composite productivity of radio for local news to one. All remaining parameters are identified relative to this convention.

4.2 An MLE Estimator Based on a Combination of Quantitative and Qualitative Time Use Data

Modeling the Categorical Time-Use Data

Consider the problem of modeling the qualitative time-use data that characterize the time allocated to different news types. Let us define the continuous latent variables H_j^* as:

$$\ln H_j^* = \ln H_j(S, x, w|\theta) + \epsilon_j \quad j = l, n, i \quad (25)$$

where $\ln H_j(S, x, w|\theta)$ is obtained from equation (19). The error term ϵ_j can be interpreted as ex-post shocks to the time allocations realized after the decision problem above has been solved. Alternatively, the error may reflect differences in how survey participants interpret and answer the survey questions.¹³ We assume that these errors follow a logistic distribution with a common scale parameter σ_ϵ that does not depend on j . Since the responses are categorical, they identify how news time varies across respondents, but not its overall level: shifting the latent index and all cut-points by the same amount leaves the conditional choice probabilities unchanged, so the categorical responses alone cannot pin down the magnitude of news time in minutes. We therefore anchor the level of H_j with the quantitative time-use data, adding moments that match average news time in the data, as discussed below.

Define the observed random variables H_j^o such that they reflect the answers to the

¹³We treat this component as independent of the underlying time allocations H_j ; response styles that varied systematically with actual news use would confound the two.

survey question on how closely the individuals follow each news type.¹⁴ There are four categorical answers:

1. Not at all closely: $H_j^o = 0$ if $H_j^* \leq \bar{H}_l$
2. Not very closely: $H_j^o = 1$ if $\bar{H}_l < H_j^* \leq \bar{H}_m$,
3. Somewhat closely: $H_j^o = 2$ if $\bar{H}_m < H_j^* \leq \bar{H}_h$
4. Very closely: $H_j^o = 3$ if $\bar{H}_h < H_j^*$.

Note that the threshold values $(\bar{H}_l, \bar{H}_m, \bar{H}_h)$ do not depend on j . This restriction guarantees that the three indices are comparable and on the same scale. Integrating out the error terms, we obtain the standard ordered logit probabilities. It should be clear from the analysis above that we need to assume that the scales used by individuals for expressing the relative importance of the different sources and types of news are the same, with the same latent variable thresholds for comparisons of local, national, and international news.¹⁵

Similarly, consider the time allocation problem among news providers. Note that we only observe these variables for the time allocated to local news in our survey, and not for national or international news. Define latent variables h_{ls}^* as the total local news time allocated to provider s :

$$\ln h_{ls}^* = \ln h_{ls}(S, x, w|\theta) + \nu_{ls} \quad s = 1, \dots, |S| \quad (26)$$

where $h_{ls}(S, x, w|\theta)$ is obtained by substituting equation (19) into equation (8). Again, the error ν_{ls} captures ex-post shocks to the time allocation problem and idiosyncratic

¹⁴The closeness question may capture attention intensity as well as time spent. However, this question is used only at the broad news-type level, where a monotone relationship between closeness and time is plausible. At the provider level, we rely on the more neutral “How often do you get local news and information from ...?” question, which maps more directly to time use.

¹⁵Common thresholds are the standard assumption in ordered choice models (Andrich, 1978; Greene and Hensher, 2010) and are necessary for identification from ordinal data alone; richer cardinal anchors, such as average time by provider, would allow provider-specific thresholds. In our setting, all three news types use the identical response format and verbal anchors, making this assumption natural.

differences in responses to survey questions. As before, we assume that ν_{ls} follows a logistic distribution with a location parameter of 0 and a scale parameter σ_ν that is common across providers.¹⁶

Recall that the survey asks the question: “How often do you get local news and information from each of the following types of sources?” The answer is a categorical variable, denoted by h_{ls}^o , that takes four values. To map this variable into our model, we assume that

1. Never: $h_{ls}^o = 0$ if s is not in the chosen bundle.¹⁷
2. Hardly ever: $h_{ls}^o = 1$ if $h_{ls}^* \leq \bar{h}_l$,
3. Sometimes: $h_{ls}^o = 2$ if $\bar{h}_l < h_{ls}^* \leq \bar{h}_h$
4. Often: $h_{ls}^o = 3$ if $\bar{h}_h < h_{ls}^*$.

Again, the thresholds do not depend on s . This restriction makes sure that the indices are comparable and on the same scale. Integrating out the error terms, we obtain the conditional choice probabilities. Note that these categorical variables are particularly informative about the productivity of each provider.

We have a random sample of size N . We assume that the errors are independently distributed across individuals n . The likelihood function of observing the two types of categorical variables can be written as

$$\begin{aligned}
 L^N(\theta) &= \prod_{n=1}^N \prod_j \prod_{k=0}^3 P_\theta(H_{nj}^o = k | S_n, x_n, w_n)^{1\{k \text{ observed}\}} \\
 &\times \prod_{n=1}^N \prod_{s \in S_n} \prod_{h=1}^3 P_\theta(h_{nls}^o = h | S_n, x_n, w_n)^{\mathbb{I}(h \text{ is observed, } s \text{ in } S_n)}
 \end{aligned} \tag{27}$$

¹⁶Specifying the measurement equations in logs makes the shocks multiplicative in levels: a shock moves time use by the same proportion for heavy and light users, and the implied time allocations are positive for any realization of the shock.

¹⁷Treating a “never” response as absence of the provider from the respondent’s bundle is one modeling choice. An alternative is to add a separate threshold for “never” and treat all five providers as potentially available to every respondent.

The first term captures the likelihood of the time allocated to the three news types. The second term captures the time for local news allocated to each provider in the choice set.

Adding Constraints based on Quantitative Time-Use Data

As discussed above, the categorical responses identify how time use varies across respondents but not its level. To anchor the level, we add moments based on the quantitative survey data from the MCS to the objective function. Recall that the MCS provides the conditional means of the total time allocated to news, conditional on age, as shown in Table 5. The optimal time-use $H(S, x, w|\theta)$ is given by equation (18). As a consequence, we can form additional moments of the form:

$$\frac{1}{N} \sum_{n=1}^N [H_n - H(S_n, x_n, w_n|\theta)] \quad (28)$$

for different age groups. Time-use is measured in minutes per day in the MCS, so these moments anchor the level of news time in minutes.

We use these moments to define a penalty function. Adding the penalty function to the likelihood function, we obtain the following objective function:

$$\ln L_N^P(\theta) = \ln L^N(\theta) - \lambda \sum_{g=1}^G \left(\frac{\bar{H}_g - \bar{H}_g(\theta)}{\bar{H}_g} \right)^2 \quad (29)$$

where $\bar{H}_g$ is the mean time allocated to news by age group g in the MCS, $\bar{H}_g(\theta)$ is its model counterpart, the average of $H(S_n, x_n, w_n|\theta)$ over the individuals in age group g , and λ is a penalty parameter. Our estimator of the parameters of the model then maximizes the penalized likelihood function. We have assumed that the error terms are independent across the different responses of an individual, conditional on observables. Appendix B relaxes this assumption by introducing unobserved heterogeneity in the taste for local news, which correlates an individual's local news responses.

Note that this estimator builds on [Imbens and Lancaster \(1994\)](#), who proposed

to combine micro and aggregate data in a constrained MLE framework. While they are primarily concerned with increasing the efficiency of the estimator, we combine the different data to anchor the level of time use. Moreover, the moments that we add are nonlinear in the parameters, which makes implementing a constrained MLE estimator difficult.¹⁸ Finally, we would like to point out that some distributional assumptions regarding the error term could be relaxed by adopting semi-parametric discrete choice models.

4.3 Adding the Stated-Preference Data to the Estimation

Next, we discuss how to integrate the stated preference data into the estimation strategy. Recall that our survey also elicits data on the valuations of the different news topics. To match the model to the data, define another latent variable

$$\ln U_l^* = \ln U_l(S, x, w) + \eta_l \quad (30)$$

where $U_l(S, x, w)$ is obtained by substituting equation (19) into equation (10). We assume that the error term η_l follows a logistic distribution with a location parameter of 0 and scale parameter σ_η . Recall that the survey asks “How important is it for you to know about each of the following topics?” for eleven local news topics, each on a four-point scale. We aggregate the responses by averaging the eleven topic ratings and discretizing the average into four ordered categories of local news importance, and denote by U_l^o the ordered category, indexed by $m = 1, \dots, 4$. The observed category is determined by cut-points on the latent variable:

1. $U_l^o = 1$ if $\ln U_l^* \leq c_1$,
2. $U_l^o = 2$ if $c_1 < \ln U_l^* \leq c_2$,
3. $U_l^o = 3$ if $c_2 < \ln U_l^* \leq c_3$,
4. $U_l^o = 4$ if $c_3 < \ln U_l^*$.

¹⁸Instead of using a penalized likelihood estimator, we could use a GMM estimator, which stacks the moments associated with the score of the likelihood function and the moments obtained from the MCS.

As before, integrating out the error term, we obtain the standard ordered logit probabilities. The three cut-points $c_1 < c_2 < c_3$ and the scale σ_η are estimated jointly with the structural parameters; no element of the measurement model is fixed a priori.

Adding the stated preference block to the likelihood function in equation (27) yields

$$\tilde{L}^N(\theta) = L^N(\theta) \times \prod_{n=1}^N \prod_{m=1}^4 P_\theta(U_{nl}^o = m \mid S_n, x_n, w_n)^{1\{m \text{ observed}\}} \quad (31)$$

and the corresponding penalized objective function is

$$\ln \tilde{L}_N^P(\theta) = \ln \tilde{L}^N(\theta) - \lambda \sum_{g=1}^G \left(\frac{\bar{H}_g - \bar{H}_g(\theta)}{\bar{H}_g} \right)^2 \quad (32)$$

with the same penalty terms as in equation (29). The observed variation in the index across individuals is particularly useful to identify the preference weights $\gamma_l(x)$ on local news.

4.4 A Monte Carlo Experiment

The previous subsections define four estimators that differ in the data made available to them: the first uses complete time diaries at the news type and provider level, the second uses only the marginal totals by type and by provider, the third replaces the quantitative measures with the ordinal survey responses anchored by the aggregate moments, and the fourth adds the stated importance index. All four are designed to estimate the parameters of the same model under slightly different assumptions. The Monte Carlo experiment serves two purposes. First, it verifies that each data configuration identifies the parameters of the model and quantifies the precision that can be expected at sample sizes comparable to our application. Second, it isolates the informational content of each data source: how much is lost when complete time diaries are replaced by marginal totals, how much of that loss is recovered by ordinal survey responses combined with a small set of aggregate anchors, and what the stated preference questions add. The answers guide the interpretation of the empirical

estimates and inform the design of surveys intended to support the estimation of time allocation models.

For the Monte Carlo estimation, we set the parameter values so that the simulated data mimic the data used in the application. As in the application, we impose the logarithmic specification of the news production function. We hold the demographic and socioeconomic characteristics, wages, and provider bundles of the $N = 27,352$ individuals in our estimation sample fixed at their observed values, set the parameters to values close to our empirical estimates, and simulate from the model the survey responses that each estimator observes. Each experiment consists of B independent replications; in each we draw a new set of responses and reestimate the model with the same estimator that is used in the application. For every parameter we compute the *bias*, the average across replications of the percent deviation of the estimate from its true value, and the *root mean squared error*, the square root of the average squared percent deviation; when the bias is negligible, the latter is the standard deviation of the estimator in percent of the truth. Table 6 collects the parameters into groups, the opportunity cost of time, the productivity composites, the demographic shifters, and the measurement thresholds and scales, and reports the median of each statistic across the parameters of a group; the median is robust to the large percent deviations that arise for coefficients with true values close to zero. Panel B contains the finite sample results.

In addition to the finite sample experiments, we evaluate each estimator at the population level by replacing every individual’s simulated categorical response with the response probabilities implied by the model at the true parameter values. The resulting objective is the expected penalized log likelihood: it contains no sampling noise and is maximized at the truth whenever the model is identified. We reestimate the model on this sampling-noise-free objective with the same optimizer as in the finite sample experiments, so a failure to recover the true parameters reflects a property of the model and the data configuration rather than sampling error. Panel A reports the resulting *median absolute error*, the median across each parameter group of the absolute percent deviation of the recovered value from its true value; a value near zero means the estimator returns the truth once sampling noise is switched off. Table 6

reports the results.

Quantitative time use data (Estimators 1 and 2). With continuous time allocations, whether at the cell level or as marginal totals, all parameters are recovered essentially without error: the demographic preference coefficients are matched almost exactly, and $\hat{\beta}$ has a bias of +0.03 and +0.09 percent with root mean squared errors of 0.53 and 0.89 percent. The information lost by aggregating the complete diaries to marginal totals is therefore minimal.

Qualitative time use data (Estimator 3). The third estimator maximizes the penalized likelihood function of equation (29), using the ordinal responses and the MCS anchors. The demographic gradients remain well identified by the variation across individuals, but the precision on the opportunity cost of time drops: $\hat{\beta}$ has a bias of +1.01 percent and a root mean squared error of 5.51 percent.

Adding stated preference data (Estimator 4). The fourth estimator adds the stated importance index and maximizes the penalized likelihood function of equation (32). The stated responses discipline the parameters through a channel that is distinct from the time use data: they depend on the opportunity cost of time through the level of the utility index rather than through the allocation of time. $\hat{\beta}$ is recovered with a bias of +0.63 percent and a root mean squared error of 2.24 percent, well below the 5.51 percent of the ordinal time-use estimator: the stated responses restore much of the precision lost when continuous diaries are replaced by ordinal categories.

In summary, we offer two conclusions. First, the parameters of the model are recovered accurately under every data configuration, including the configuration of our empirical application in which only ordinal time use responses and a small set of aggregate anchors are available. Second, the precision of the estimator depends on the richness of the revealed-preference data. Complete diaries recover the parameters most precisely, marginal totals less so, and the ordinal survey responses least precisely (Estimators 1 through 3). The stated-importance index used in Estimator 4 improves upon Estimator 3 by roughly halving the root mean squared errors of the ordinal

estimator and narrowing the gap to the precision of continuous time diaries. As such, incorporating stated preference data into the analysis can help improve the precision of the estimator.

Table 6: Monte Carlo Results by Data Type

Parameters	Statistic	Estimator			
		1	2	3	4
<i>Panel A: population level (median absolute error, percent)</i>					
β	med. err , %	0.00	0.05	0.02	0.00
t_{js}	med. err , %	0.00	0.05	0.00	0.00
$\gamma_j(x)$	med. err , %	0.00	0.01	0.01	0.00
$\bar{H}, \bar{h}, \sigma_\epsilon, \sigma_\nu$	med. err , %	—	—	0.01	0.00
c_m, σ_η	med. err , %	—	—	—	0.00
<i>Panel B: finite sample (median bias and median RMSE, percent)</i>					
β	bias, %	+0.03	+0.09	+1.01	+0.63
	RMSE, %	0.53	0.89	5.51	2.24
t_{js}	bias, %	−0.00	−0.06	−0.08	−0.34
	RMSE, %	0.76	1.37	8.55	4.55
$\gamma_j(x)$	bias, %	+0.00	−0.01	−0.10	−0.06
	RMSE, %	0.69	1.04	9.89	4.97
$\bar{H}, \bar{h}, \sigma_\epsilon, \sigma_\nu$	bias, %	—	—	−0.01	−0.05
	RMSE, %	—	—	2.37	2.21
c_m, σ_η	bias, %	—	—	—	−0.09
	RMSE, %	—	—	—	0.78
Replications B		300	300	300	300

Notes: The data generating process holds the parameters at the empirical estimates of Section 5. Estimator 1 observes complete time diaries; Estimator 2 marginal time allocations, both with log normal measurement error of standard deviation 0.3; Estimator 3 the ordinal time use responses with the MCS anchors; Estimator 4 adds the stated importance index. Entries are medians across the parameters of each group, in percent of the true values: the median absolute error at the population objective in Panel A, and the median bias and root mean squared error across the B replications in Panel B.

5 Empirical Results

We have estimated several specifications of our model using the survey data reported in Section 2. Our preferred model is relatively parsimonious; it has fifteen composite productivities, of which fourteen are estimated and one is fixed by the normalization, twenty-four parameters that capture heterogeneity in preferences, the parameter that

captures the opportunity costs of time (β), and a set of nuisance parameters, namely the thresholds and scales of the ordered response models for time use and the cut-points and scale of the stated importance index. Overall, we find that our model fits the observed data rather well. Appendix C compares the observed and predicted distributions of the categorical responses.

5.1 Preferences

Table 7 reports the parameter estimates and estimated standard errors for the parameters that characterize heterogeneity in preferences for the three news types.¹⁹

Table 7: Preference Parameters

Variable	Local	National	International
log(Income)	0.21 (0.02)	0.04 (0.03)	0.06 (0.02)
Age 25-34	-0.22 (0.03)	-1.12 (0.06)	-2.17 (0.09)
Age 35-54	-0.28 (0.05)	-0.19 (0.05)	-1.54 (0.07)
Age 55-65	-0.22 (0.02)	-0.06 (0.02)	-0.57 (0.04)
Male	-0.17 (0.03)	0.91 (0.04)	1.14 (0.05)
African American	0.96 (0.05)	-0.10 (0.07)	-0.11 (0.07)
Hispanic	0.64 (0.04)	0.37 (0.06)	0.71 (0.06)
College Grad	0.14 (0.02)	1.01 (0.04)	1.04 (0.04)

Notes: Penalized maximum likelihood estimates (Estimator 4), LNS estimation sample ($N = 27,352$). Estimated standard errors in parentheses.

We find substantial heterogeneity in preferences for the three news types by race, ethnicity, age, gender, and education. Preferences tend to increase with income.

¹⁹Our estimate of β is 0.25 with a standard error of 0.005. The parameter estimates and standard errors of the nuisance parameters are available upon request from the authors.

Males have stronger preferences for national and international news than females. In addition, there is important heterogeneity associated with education. College-educated individuals have stronger preferences for all types of news than non-college-educated individuals. These differences are most pronounced for national and international news.

An important finding is that minority individuals typically have stronger preferences for local news than white individuals. The stated importance responses of Section 2 show that these differences extend to almost all local news topics covered in the survey, and are most pronounced for economically salient topics. In contrast, white individuals have stronger preferences for national and international news than African Americans. Notably, Hispanics have stronger preferences for national and international news than whites. These findings are consistent with recent research in labor and urban economics, which has also documented that minority and non-college-educated individuals are more heavily exposed to shocks to the local economy than white and college-educated individuals. In particular, they have lower mobility rates, are more strongly exposed to shocks in the local labor market, rely more heavily upon informal networks for job referrals, have fewer options in the local housing markets, and are more likely to be affected by shocks in neighborhood amenities such as crime and public school quality than other individuals.²⁰ Since minority and non-college-educated individuals are more exposed to local shocks, it is natural to expect them to pay closer attention to changes in the local environment than white and college-educated individuals. Our findings are consistent with this hypothesis.²¹

We have argued that it makes basic economic sense that African Americans and Hispanics allocate more time to local news than white Americans. A direct test

²⁰See, for example, [Altonji and Blank \(1999\)](#), [Shuey and Willson \(2008\)](#), [Hoynes et al. \(2012\)](#), and [Bayer et al. \(2016\)](#).

²¹Research in labor economics has also emphasized the importance of informal networks in labor markets, especially for younger, non-college-educated, male workers. [Ioannides and Loury \(2004\)](#) and [Bayer et al. \(2008\)](#) highlight neighborhood referrals and assortative matching in social networks. [Bailey et al. \(2020\)](#) analyze data from Facebook to explore the spatial structure of social networks in the New York metro area. They find that a substantial share of urban residents' connections is to individuals who are located nearby. That suggests that even in the digital economy, most information about the availability and suitability of local jobs is propagated via online social networks. We also find that individuals rely on a variety of formal and informal news outlets to stay informed.

would require quantitative time diaries for news, which the ATUS does not collect. The closest activity the ATUS does cover is television watched for entertainment, and the patterns line up: African Americans watch significantly more television for entertainment in the ATUS, just as they consume more television news in the LNS, and the same holds for non-college-educated individuals. The racial and education differences in the LNS are thus consistent with the time use patterns the ATUS can measure. Appendix D provides the details.

5.2 News Production Functions

Table 8 reports the parameter estimates and estimated standard errors for the composite productivities t_{js} , one for each combination of news type and provider. The composite for local news on radio is fixed by the normalization discussed in Section 4. All other composites are measured relative to it.

Table 8: News Production Function Parameters

	Local	National	International
Newspaper	1.640 (0.078)	4.362 (0.473)	1.188 (0.146)
TV	12.268 (0.770)	2.593 (0.401)	0.634 (0.116)
Radio	1.000 [†] —	0.775 (0.296)	0.276 (0.089)
Online	4.506 (0.256)	9.793 (0.826)	3.000 (0.307)
Social Network	1.197 (0.041)	0.098 (0.031)	0.095 (0.028)

Notes: Penalized maximum likelihood estimates (Estimator 4). Estimated standard errors in parentheses. [†]Fixed by the normalization.

The estimates show that provider productivity differs sharply across news types. Television is by far the most productive provider for local news, but substantially less so for national and international news. For those types the ranking reverses: online platforms are the most productive provider, followed by printed newspapers, which are considerably more productive for national than for local news. Social media

contributes little outside local news, and radio is the least productive provider for every type. The concavity of the production function rationalizes the observation that most individuals obtain news from multiple sources.

The model links time to money. It recovers news consumption in minutes and, through the estimated opportunity cost of time, values this time in dollars. Neither quantity is directly observed in our data, and both are recovered from the categorical responses and a small set of aggregate anchors. A natural illustration of this link is to ask how much individuals are willing to pay for improvements in the news production function. We perform this exercise separately for local, national, and international news. More specifically, we increase the composites t_{js} , which capture the productivity of providers for each news type, by ten percent and then compute the willingness of each individual in our sample for such an increase in productivity. Table 9 summarizes our main findings.

Overall, we find that our WTP estimates are plausible and generate important new insights into the distribution of welfare effects associated with changes in the productivity of news provision. In particular, we find that a ten percent increase in the productivity of local news leads to an average increase of 3.1 minutes per day of local news consumption, which is valued at \$3.57 per day. Similarly, a ten percent increase in the productivity of national news is valued at \$4.75 per day. Finally, a ten percent increase in the productivity of international news is valued at \$0.86 a day.²² These willingness-to-pay estimates exceed the simple back-of-the-envelope valuation that multiplies the increase in the time allocated to news by the wage rate, because the productivity improvement also raises the value of the infra-marginal time already devoted to news.²³

²²An alternative approach for measuring the valuation of news is to design and implement a survey experiment to generate direct evidence on how people select, acquire and process information, as done by Fuster et al. (2022).

²³These daily willingness to pay estimates can be compared to simple back-of-the-envelope calculations that multiply the predicted daily time changes by the average wage rate. Such calculations would suggest daily valuations of \$0.98, \$1.23, and \$0.32 for local, national, and international news, respectively. As a benchmark, daily subscription prices for major newspapers are typically between \$0.50 and \$1.50 per day, and basic cable television packages cost roughly \$2.00 to \$3.00 per day. Our WTP estimates for a ten percent productivity improvement are within the range of these daily expenditures on news access.

Table 9: Willingness to Pay for a 10 Percent Productivity Increase

	Local	National	International
<i>Willingness to pay, dollars per day</i>			
Overall	3.57	4.75	0.86
Age 18-29	2.73	1.05	0.06
Age 30-49	3.08	4.29	0.20
Age 50-64	3.65	5.10	0.81
Age 65 or above	4.22	5.95	1.84
White	2.79	4.82	0.86
African American	8.96	3.14	0.42
Hispanic	5.27	5.33	1.13
HS Grad	3.39	1.97	0.32
CL Grad	3.73	7.39	1.38
Women	3.91	2.29	0.27
Men	3.11	8.01	1.65
Married	3.58	5.43	1.03
Single	3.55	3.84	0.63
Democrat	4.23	4.55	0.81
Republican	3.14	4.85	0.93
<i>Implied increase in news time, minutes per day</i>			
Overall	3.1	3.4	0.8

Notes: Model-based willingness to pay for a ten percent increase in the productivity of each news type, computed for each individual in the estimation sample and averaged within groups.

5.3 Differences in Time-Use and Informational Gaps

We now turn to the time-use patterns implied by the estimated model.²⁴ Figure 2 plots the densities of time-use allocations by race and ethnicity predicted by our model. The differences in local news time are economically meaningful. On average, African Americans spend about 74 minutes per day on local news, Hispanics 47 minutes, and whites 24 minutes. The pattern reverses for national news, where whites

²⁴As noted in Section 4, a “Never” response is treated as absence of the provider from the bundle, and we take bundles as given in the counterfactual exercises. Under the alternative interpretation, in which all providers are available and “Never” is the lowest time-use category, the variation attributed to the access channel would be absorbed by preferences, reinforcing our finding that preferences and opportunity costs are the primary drivers of informational gaps.

and Hispanics spend somewhat more time than African Americans, and time on international news is short for every group.

The size of these gaps does not reveal their source. In the model, a gap in news time between two groups can come from three factors: differences in preferences, differences in the opportunity cost of time measured by wages, and differences in access to news providers. The distinction matters because the factors carry different implications. A gap driven by wages would narrow if wage differences narrowed, while a gap driven by preferences would persist. Using the estimated model, we quantify the contribution of each factor.

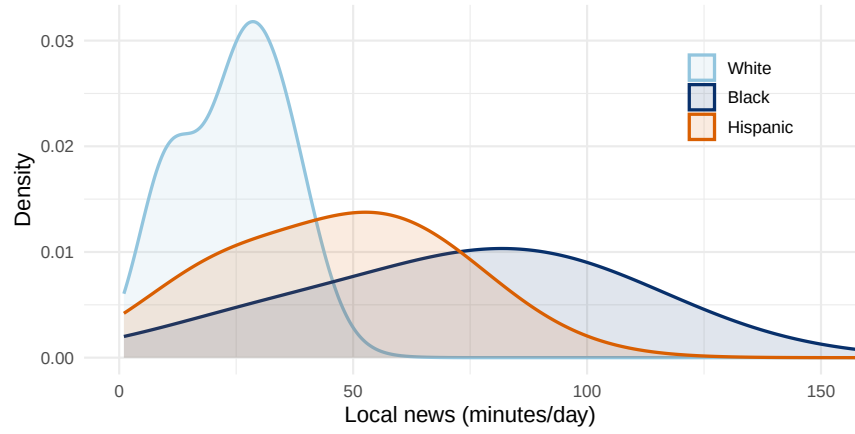
Table 10: Decomposition of Time-Use Gaps by News Type and Demographic Groups

		I	II	III	IV
News Type		Base Gap	Remove Wage Diff	Remove Pref Diff	Net Effect
African American vs White	Local	49.58	42.69	3.86	1.29
	National	-5.08	-7.58	-2.02	-4.80
	International	-2.52	-3.02	-1.88	-2.44
Hispanic vs White	Local	22.17	19.30	0.02	-1.44
	National	7.35	4.92	-5.27	-6.89
	International	2.65	1.99	-2.83	-3.14
College vs High School	Local	-5.83	3.93	-9.49	-1.14
	National	24.76	41.60	-4.26	1.46
	International	6.31	10.48	-1.02	0.34

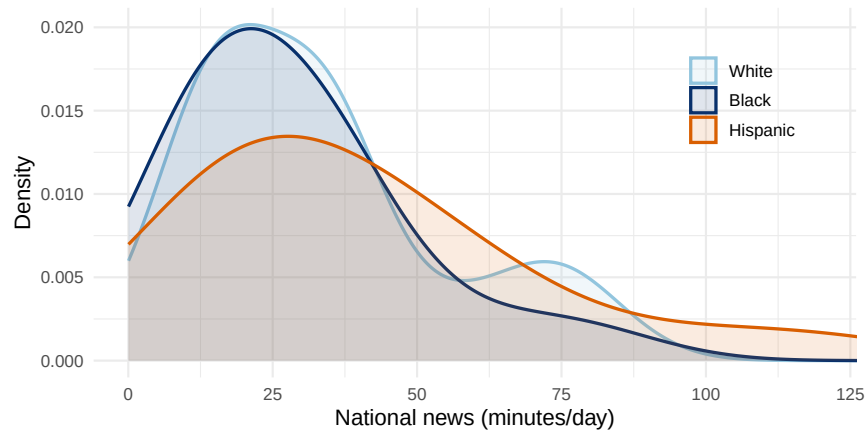
Notes: All outcomes are measured in minutes per day. Column I is the model-predicted difference in time allocations between groups. Columns II, III, and IV report the gap that remains after removing wage differences, preference differences, and both, respectively. Because the model is nonlinear, the reductions in columns II and III do not sum exactly to the reduction in column IV.

Table 10 reports the findings from the decomposition exercises. Column I is the model-predicted gap between groups. Columns II and III report the gap that remains after removing wage differences and preference differences, and Column IV the gap that remains after removing both. Column IV therefore measures the importance of differences in access to news providers. Because the model is nonlinear, the two removals interact, and the individual reductions in Columns II and III need not sum

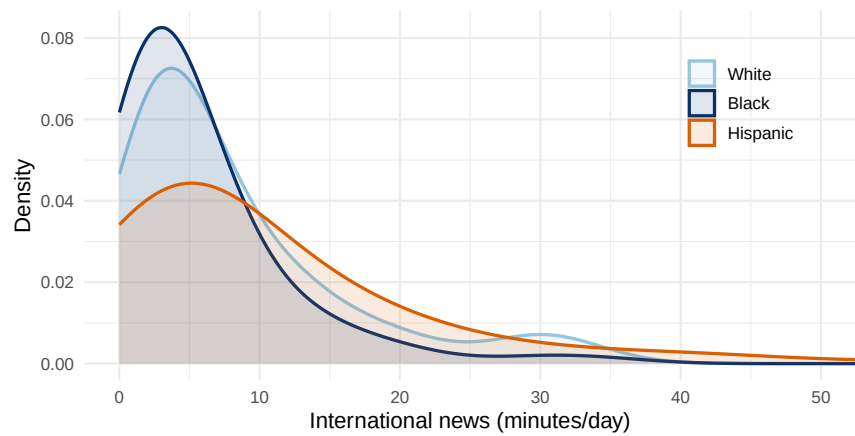
Figure 2: Predicted Time Allocations by Race and Ethnicity



(a) Local



(b) National



(c) International

Notes: Model-predicted distributions of daily³⁷ news time by race and ethnicity, simulated at the parameter estimates of Section 5.

to the joint reduction in Column IV.

Recall that the largest gap between African Americans and whites is in local news consumption, about 49 minutes per day. African Americans have stronger preferences for local news than whites, and they also have lower wages and hence lower opportunity costs of time. Differences in preferences account for the overwhelming majority of the gap: removing them shrinks the gap from about 50 to about 4 minutes per day. Removing wage differences on its own reduces the gap to about 43 minutes, so wages account for a smaller share of roughly 7 minutes. Differences in access to news providers explain essentially none of the gap.

The composition of the local news gap is similar for Hispanics, though even more concentrated in preferences: removing preference differences essentially eliminates the 22 minute gap, while removing wage differences reduces it by about 3 minutes. As for African Americans, differences in access to providers explain essentially none of the gap. Hispanics also consume somewhat more national and international news than whites, differences driven mainly by preferences.

We have done a similar analysis by education. We find that there are large and significant differences in local news consumption. On average, non-college-educated individuals spend 34 minutes per day on local news, while college-educated individuals spend 28 minutes. In contrast, college-educated individuals spend considerably more time on national (21 versus 46 minutes) and international news (5 versus 11 minutes). Again, this finding is consistent with research in labor economics that non-college-educated individuals are more exposed to shocks in the local economy and rely more heavily on local referrals to obtain jobs. College-educated individuals tend to participate in regional or national labor markets.

Table 10 shows the decomposition of the educational gaps. Here we find that preferences and wage effects go in opposite directions. While college-educated individuals have stronger preferences for all news types, they have higher wages and hence higher opportunity costs of time. These two effects tend to offset each other. Table 10 shows that the wage effect tends to dominate the preference effect for local news, while the preference effect dominates the wage effect for national and international news. Differences in access to news providers are not important.

In summary, we have shown that the observed differences in informational gaps for all news types are driven by differences in preferences, opportunity costs of time (wages), and access to or adoption of news providers. We find that the first two channels matter the most, i.e., differences in access to news providers only explain a small fraction of the observed informational gaps. There is not much variation in the choice sets in our sample. More than 60% individuals have access to 4 or 5 providers in their available bundles. Individuals with only one provider in the choice set are less than 3 percent of the sample. There are also only negligible differences by race or education. For example, African Americans have, on average, 3.86 providers in their choice set while whites have 3.78. In contrast, minorities (African Americans and Hispanics) have, on average, significantly lower opportunity costs of time and stronger preferences for local news than whites. Each factor explains a significant fraction of the differences in predicted time allocations to local news. For national and international news, whites have stronger preferences than African Americans, and these are partially offset by their higher opportunity costs of time. The differences in time allocations by education follow a similar pattern. Wage effects dominate for local news, while preference effects are most salient for national and international news.

Appendix [B](#) discusses an extension that allows for unobserved heterogeneity in the taste for local news and shows that the model remains identified.

6 Conclusions

This paper has developed a maximum likelihood estimator that exploits both quantitative time-use data and qualitative survey information to estimate a model of time allocation with endogenous information acquisition. We show that the qualitative survey responses can be treated as noisy but informative measurements of latent economic objects, such as preferences and time allocations, and thus interpreted within the context of our model. By embedding these measurements in a likelihood framework, the survey responses provide information beyond that in standard choice data. This additional information helps separately identify preferences and production tech-

nology, which choice data alone cannot pin down when key variables are unobserved or imperfectly measured. More broadly, the framework demonstrates how survey data can be useful while acknowledging that the data environment is characterized by incomplete or imprecise measurement. Our Monte Carlo exercise compares these estimators to more traditional estimators based on ideal time diary data. We find that our estimators perform admirably well compared to these benchmarks.

We then apply this approach to study time allocations to local, national and international news. We document substantial disparities in news consumption by race, ethnicity, and education. Non-college-educated and minority individuals devote more attention to local news, while college-educated and white individuals allocate relatively more time to national and international news. Decomposing these gaps, we find that the local news gap is driven by both lower wages and stronger local news preferences among minorities, whereas the national and international gaps are primarily preference-driven, with opportunity costs partially offsetting them. These patterns align with existing evidence on mobility, local exposure, and time use, providing external validation for the survey-based measures.

Although we focus on news acquisition, our approach applies wherever a detailed qualitative survey records the frequency of specific behaviors and a coarser quantitative source anchors their scale. Two examples are parental time investment, where developmental surveys record how often parents engage in activities while time-use data report only total childcare time, and financial literacy, where surveys record ordinal financial behaviors alongside coarse anchors on wealth or savings. In each case, the qualitative data identify how latent outcomes vary across individuals, while the quantitative moments fix the scale at which these comparisons should be interpreted.

References

- Aguiar, Mark and Erik Hurst**, “Measuring Trends in Leisure: The Allocation of Time over Five Decades,” *The Quarterly Journal of Economics*, 2007, *122* (3), 969–1006.
- **and —**, “The Macroeconomics of Time Allocation,” in John B. Taylor and Harald Uhlig, eds., *Handbook of Macroeconomics*, Vol. 2, Elsevier, 2016, pp. 203–253.
- Almås, Ingvild, Orazio Attanasio, and Pamela Jervis**, “Presidential Address: Economics and Measurement: New Measures to Model Decision Making,” *Econometrica*, 2024, *92* (4), 947–978.
- Altonji, Joseph G and Rebecca M Blank**, “Race and gender in the labor market,” *Handbook of Labor Economics*, 1999, *3*, 3143–3259.
- Andrich, David**, “A Rating Formulation for Ordered Response Categories,” *Psychometrika*, 1978, *43*, 561–573.
- Angrist, Joshua D. and Alan B. Krueger**, “The Effect of Age at School Entry on Educational Attainment: An Application of Instrumental Variables with Moments from Two Samples,” *Journal of the American Statistical Association*, 1992, *87* (418), 328–336.
- Arcidiacono, Peter, V. Joseph Hotz, and Songman Kang**, “Modeling College Major Choices Using Elicited Measures of Expectations and Counterfactuals,” *Journal of Econometrics*, 2012, *166* (1), 3–16.
- Arrow, Kenneth J**, “Rational choice functions and orderings,” *Economica*, 1959, *26* (102), 121–127.
- Bailey, Michael, Patrick Farrell, Theresa Kuchler, and Johannes Stroebel**, “Social Connectedness in Urban Areas,” *The Journal of Urban Economics*, 2020, *118*, 103264.

- Bastian, Jacob and Lance Lochner**, “The Earned Income Tax Credit and Maternal Time Use: More Time Working and Less Time with Kids?,” *Journal of Labor Economics*, 2022, 40 (3), 573–611.
- Bayer, Patrick, Fernando Ferreira, and Stephen L. Ross**, “The Vulnerability of Minority Homeowners in the Housing Boom and Bust,” *American Economic Journal: Economic Policy*, February 2016, 8 (1), 1–27.
- , **Stephen Ross, and Giorgio Topa**, “Place of work and place of residence: Informal hiring networks and labor market outcomes,” *The Journal of Political Economy*, 2008, 116 (6), 1150–96.
- Becker, Gary S**, “A Theory of the Allocation of Time,” *The Economic Journal*, 1965, 75 (299), 493–517.
- Berry, Steven, James Levinsohn, and Ariel Pakes**, “Differentiated Products Demand Systems from a Combination of Micro and Macro Data: The New Car Market,” *Journal of Political Economy*, 2004, 112 (1), 68–105.
- Biddle, Jeff E and Daniel S Hamermesh**, “Sleep and the Allocation of Time,” *Journal of Political Economy*, 1990, 98 (5), 922–943.
- Blass, Asher A., Saul Lach, and Charles F. Manski**, “Using Elicited Choice Probabilities to Estimate Random Utility Models: Preferences for Electricity Reliability,” *International Economic Review*, 2010, 51 (2), 421–440.
- Blundell, Richard, Luigi Pistaferri, and Itay Saporta-Eksten**, “Children, Time Allocation, and Consumption Insurance,” *Journal of Political Economy*, 2018, 126 (S1), S73–S115.
- Bond, Timothy N. and Kevin Lang**, “The Sad Truth about Happiness Scales,” *Journal of Political Economy*, 2019, 127 (4), 1629–1640.
- Chen, Xiaohong, Han Hong, and Alessandro Tarozi**, “Semiparametric Efficiency in GMM Models with Auxiliary Data,” *Annals of Statistics*, 2008, 36 (2), 808–843.

- Cherchye, Laurens, Bram De Rock, and Frederic Vermeulen**, “Married with Children: A Collective Labor Supply Model with Detailed Time Use and Intra-household Expenditure Information,” *American Economic Review*, 2012, *102* (7), 3377–3405.
- Chiappori, Pierre-Andre, Bernard Fortin, and Guy Lacroix**, “Marriage market, divorce legislation, and household labor supply,” *Journal of Political Economy*, 2002, *110* (1), 37–72.
- Chou, Cheng and Ruoyao Shi**, “What Time Use Surveys Can (and Cannot) Tell Us About Labor Supply,” *Journal of Applied Econometrics*, 2021, *36* (7), 917–937.
- Cunha, Flavio, Irma T. Elo, and Jennifer Culhane**, “Eliciting Maternal Expectations about the Technology of Cognitive Skill Formation,” Working Paper 19144, National Bureau of Economic Research 2013.
- , **James J. Heckman, and Susanne M. Schennach**, “Estimating the Technology of Cognitive and Noncognitive Skill Formation,” *Econometrica*, 2010, *78* (3), 883–931.
- de New, Sonja C. and Stefanie Schurer**, “Survey Response Behavior as a Proxy for Unobserved Ability: Theory and Evidence,” *Journal of Business & Economic Statistics*, 2023, *41* (1), 197–212.
- Dominitz, Jeff and Charles F Manski**, “Using Expectations Data to Study Subjective Income Expectations,” *Journal of the American Statistical Association*, 1997, *92* (439), 855–867.
- Fiorini, Mario and Michael Keane**, “How the allocation of children’s time affects cognitive and noncognitive development,” *Journal of Labor Economics*, 2014, *32* (4), 787–836.
- Fuster, Andreas, Ricardo Perez-Truglia, Mirko Wiederholt, and Basit Zafar**, “Expectations with Endogenous Information Acquisition: An Experimental Investigation,” *The Review of Economics and Statistics*, 2022, *104* (5), 1059–1078.

- Gentzkow, Matthew and Jesse M. Shapiro**, “Ideological Segregation Online and Offline,” *The Quarterly Journal of Economics*, 2011, *126* (4), 1799–1839.
- Ghez, Gilbert and Gary S Becker**, *The Allocation of Time and Goods over the Life Cycle*, New York: Columbia University Press, 1975.
- Greene, William H. and David A. Hensher**, *Modeling Ordered Choices: A Primer*, Cambridge University Press, 2010.
- Gronau, Reuben**, “Leisure, Home Production, and Work—the Theory of the Allocation of Time Revisited,” *Journal of Political Economy*, 1977, *85* (6), 1099–1123.
- Hoynes, Hilary, Douglas L. Miller, and Jessamyn Schaller**, “Who Suffers during Recessions?,” *Journal of Economic Perspectives*, September 2012, *26* (3), 27–48.
- Hwang, Yujung**, “Bounding Omitted Variable Bias Using Auxiliary Data: With an Application to Estimate Neighborhood Effects,” *Journal of Business & Economic Statistics*, 2026. forthcoming.
- Imbens, Guido and Tony Lancaster**, “Combining Micro and Macro Data in Microeconomic Models,” *Review of Economic Studies*, 1994, *61* (4), 655–680.
- Ioannides, Yannis and Linda Datcher Loury**, “Job information networks, neighborhood effects, and inequality,” *The Journal of Economic Literature*, 2004, *42* (4), 1056–93.
- Juster, F Thomas and Frank P Stafford, eds**, *Time, Goods, and Well-Being*, Ann Arbor: Survey Research Center, Institute for Social Research, University of Michigan, 1985.
- Kapteyn, Arie, James P. Smith, and Arthur van Soest**, “Vignettes and Self-Reports of Work Disability in the United States and the Netherlands,” *American Economic Review*, 2007, *97* (1), 461–473.
- Kerkhofs, Marcel and Peter Kooreman**, “Identification and Estimation of a Class of Household Production Models,” *Journal of Applied Econometrics*, 2003, *18* (3), 337–369.

- Kooreman, Peter and Arie Kapteyn**, “A Disaggregated Analysis of the Allocation of Time within the Household,” *Journal of Political Economy*, 1987, 95 (2), 223–249.
- Levy, Ro’ee**, “Social Media, News Consumption, and Polarization: Evidence from a Field Experiment,” *American Economic Review*, 2021, 111 (3), 831–870.
- Lise, Jeremy and Ken Yamada**, “Household Sharing and Commitment: Evidence from Panel Data on Individual Expenditures and Time Use,” *Review of Economic Studies*, 2019, 86 (5), 2184–2219.
- Maestas, Nicole, Kathleen J Mullen, David Powell, Till von Wachter, and Jeffrey B Wenger**, “The Value of Working Conditions in the United States and the Implications for the Structure of Wages,” *American Economic Review*, 2023, 113 (7), 2007–2047.
- Manski, Charles F.**, “Measuring expectations,” *Econometrica*, 2004, 72 (5), 1329–1376.
- McCullagh, Peter**, “Regression Models for Ordinal Data,” *Journal of the Royal Statistical Society, Series B*, 1980, 42 (2), 109–142.
- Munro, Evan and Serena Ng**, “Latent Dirichlet Analysis of Categorical Survey Responses,” *Journal of Business & Economic Statistics*, 2022, 40 (1), 256–271.
- Oparina, Ekaterina and Sorawoot Srisuma**, “Analyzing Subjective Well-Being Data with Misclassification,” *Journal of Business & Economic Statistics*, 2022, 40 (2), 730–743.
- Pew Research Center**, “Older Americans, Black Adults and Americans with Less Education More Interested in Local News,” 2019. August 14, 2019.
- Prior, Markus**, *Post-Broadcast Democracy: How Media Choice Increases Inequality in Political Involvement and Polarizes Elections*, Cambridge University Press, 2007.
- Ridder, Geert and Robert Moffitt**, “The Econometrics of Data Combination,” in James J. Heckman and Edward E. Leamer, eds., *Handbook of Econometrics*, Vol. 6B, Elsevier, 2007, pp. 5469–5547.

- Rogerson, Richard and Johanna Wallenius**, “Household time use among older couples: Evidence and implications for labor supply parameters,” *Quarterly Journal of Economics*, 2019, *134* (2), 1079–1120.
- Samuelson, Paul A.**, “A Note on the Pure Theory of Consumer’s Behaviour,” *Economica*, 1938, *5* (17), 61–71.
- , “Consumption Theory in Terms of Revealed Preference,” *Economica*, 1948, *15* (60), 243–253.
- Shuey, Kim M. and Andrea E. Willson**, “Cumulative Disadvantage and Black-White Disparities in Life-Course Health Trajectories,” *Research on Aging*, 2008, *30* (2), 200–225.
- Stantcheva, Stefanie**, “How to Run Surveys: A Guide to Creating Your Own Identifying Variation and Revealing the Invisible,” *Annual Review of Economics*, 2023, *15* (1), 205–234.
- van der Klaauw, Wilbert**, “On the Use of Expectations Data in Estimating Structural Dynamic Choice Models,” *Journal of Labor Economics*, 2012, *30* (3), 521–554.
- Wiswall, Matthew and Basit Zafar**, “Preference for the Workplace, Investment in Human Capital, and Gender,” *The Quarterly Journal of Economics*, 2018, *133* (1), 457–507.

A A Generalized Cross-Type Aggregator

The model in Section 3 aggregates utility across news types additively (equation (13)), so that news types are treated as perfect substitutes. This appendix shows that this assumption can be relaxed to a constant-elasticity-of-substitution (CES) aggregator with its own substitution elasticity, that the optimal time allocation still has a closed form, and that the additional parameter is separately identified from the rest of the model. The additive model is the special case in which types are perfect substitutes.

The within-type problem is unchanged. For a given time budget H_j , the optimal provider allocation and the resulting indirect utility are still given by equations (8) and (10),

$$U_j(S, x, H_j) = u_j(S, x) \frac{1}{1-\rho} H_j^{1-\rho}, \quad u_j(S, x) = \gamma_j(x) \sum_{s \in S} t_{js} w_{js}(S, x)^{1-\rho}, \quad (33)$$

where $w_{js}(S, x) = (\gamma_j(x) t_{js})^{1/\rho} / \sum_{s' \in S} (\gamma_j(x) t_{js'})^{1/\rho}$ is the optimal provider share. Note that the additive form across providers still allows imperfect substitution among them: the share w_{js} responds to the relative productivities t_{js} with an elasticity governed by ρ . The generalization only concerns the top layer.²⁵ We replace the additive aggregation across types in equation (13) with

$$U(S, x, H) = \left(\sum_{j=1}^J U_j(S, x, H_j)^\sigma \right)^{1/\sigma}, \quad \sigma \in (0, 1], \quad (34)$$

whose elasticity of substitution across types is $1/(1-\sigma)$. At $\sigma = 1$ the aggregator is additive and reproduces the baseline.

²⁵A fully nested CES, with a separate elasticity at the provider layer, is possible but less suited to this application. Provider substitution is already governed by ρ , so a separate provider elasticity would be weakly identified; and a CES provider layer would make the within-type indirect utility linear in H_j , removing the interior solution for total news time and requiring an additional concavity parameter at the outer trade-off.

Maximizing (34) subject to $\sum_j H_j = H$ gives the cross-type allocation

$$\frac{H_j}{H} = \omega_j(S, x) = \frac{u_j(S, x)^\varepsilon}{\sum_{j'} u_{j'}(S, x)^\varepsilon}, \quad \varepsilon \equiv \frac{\sigma}{1 - \sigma(1 - \rho)}, \quad (35)$$

so a single compound elasticity $\varepsilon(\rho, \sigma)$ governs how time is split across types. Substituting back into the outer trade-off $\max_H \beta U(S, x, H) + w(1 - H)$ gives the total time allocated to news,

$$H(S, x, w) = \left(\frac{\beta}{w}\right)^{1/\rho} \left(\sum_j u_j(S, x)^\varepsilon\right)^{(1 - \sigma(1 - \rho))/(\rho\sigma)}, \quad (36)$$

and the full allocation follows as $H_j = \omega_j H$ and $h_{js} = w_{js} H_j$. All objects are closed form, so the estimator of Section 4 carries over with one additional parameter.

Setting $\sigma = 1$ gives $\varepsilon = 1/\rho$, so (35) becomes $H_j \propto u_j^{1/\rho}$ and (36) becomes $H = (\beta u/w)^{1/\rho}$ with $u = (\sum_j u_j^{1/\rho})^\rho$, recovering the model of Section 3 exactly.

The closed form also survives at $\rho = 1$, the logarithmic production function estimated in the application. Setting $\rho = 1$ gives the Cobb–Douglas provider share $w_{js} = t_{js} / \sum_{s' \in S} t_{js'}$, the type composite $u_j(S, x) = \gamma_j(x) \sum_{s \in S} t_{js}$, and $\varepsilon = \sigma$, so the cross-type allocation and the total time reduce to

$$\omega_j(S, x) = \frac{u_j(S, x)^\sigma}{\sum_{j'} u_{j'}(S, x)^\sigma}, \quad H(S, x, w) = \frac{\beta}{w} \left(\sum_j u_j(S, x)^\sigma\right)^{1/\sigma}, \quad (37)$$

with $H_j = \omega_j H$ and $h_{js} = w_{js} H_j$. As in the main text, the utility level carries a common constant $1/(1 - \rho)$ that diverges at $\rho = 1$ but does not affect behavior, and the allocation rules above are its well-defined limit. At $\sigma = 1$ they collapse to the additive logarithmic model of Section 3.

The two curvature parameters are separately identified because they act on different parts of the data. The production curvature ρ is identified from the within-type provider shares w_{js} and from the response of total news time H to the wage. The cross-type elasticity σ is identified from how strongly the split H_j across types responds to differences in u_j , both across types and across individuals with different characteristics x : a strong response implies a large ε and σ near one, a weak response

a small ε . Because ρ is pinned down by the provider and wage margins while σ is pinned down by the cross-type margin, the two are not confounded, and the identification of the remaining parameters does not rely on the additive aggregation. The same construction relaxes additivity across topics within a type. We keep the additive form in the main text as a parsimonious baseline, with one fewer curvature parameter than the CES.

Estimating the cross-type elasticity on the application yields $\hat{\sigma} = 0.49$, an elasticity of substitution across news types of about two, and it is cleanly identified. Aggregate behavior is essentially unchanged: total news time and the opportunity cost of time are nearly identical to the additive specification (Table 11), because $H = \beta u/w$ pins the product βu , total news time valued at the wage, to the data, so that β and the composites u trade off with almost no effect on the implied time use. The cross-type composition does shift: imperfect substitution compresses the allocation, so the racial differences in local news time are smaller in magnitude than under additivity, though of the same sign, and the qualitative pattern that minorities spend more time on local news than whites is preserved.

Table 11: Additive and CES specifications: implied time use

	Additive	CES
Opportunity cost of time β	0.25	0.30
Cross-type elasticity σ	1 (imposed)	0.49
Total news, minutes/day	72.9	72.8
<i>Local news, minutes/day</i>		
White	24.5	21.5
African American	74.1	49.9
Hispanic	46.7	38.2
African American – White gap	49.6	28.4
Hispanic – White gap	22.2	16.7

Notes: The additive column reports the main-text estimates; the CES column reports the cross-type CES specification of this appendix. Entries other than β and σ are mean minutes per day implied by each specification.

B Unobserved Preference Heterogeneity

This appendix shows that our approach extends naturally to a richer model with unobserved heterogeneity in tastes. Two considerations motivate the extension. First, individuals who look alike in their observed characteristics may still differ in how much they value news, and an individual random effect captures this remaining variation. Second, because the same random effect enters all of a person’s local responses, it allows these measurements to be correlated within individuals rather than independent given observables. Concretely, we extend Estimator 4 with an individual random effect in the taste for local news, $\ln \gamma_{l,i} = x'_{li}g_l + \alpha_i$, with α_i normally distributed with standard deviation σ_α . The effect shifts every local response of an individual, the closeness category, the five provider frequencies, and the stated importance category.

The extension remains identified from the same categorical data. The random effect α_i shifts all of a person’s local responses together, so it induces co-movement among them beyond what the observed characteristics and the independent measurement errors explain. The strength of this within-person co-movement identifies σ_α : were σ_α zero, the local responses would be conditionally independent given observables, so any systematic co-movement reveals the dispersion of the random effect. The stated importance index adds a further local measurement that makes this identification robust, while the level of the opportunity cost of time is still pinned by the aggregate anchors as in the main text. Estimating the richer model requires integrating over the random effect, for example by Gauss-Hermite quadrature; we leave a full estimation and simulation study to future work.

How one interprets α_i does not change its usefulness. It may capture genuine taste heterogeneity, or it may reflect an individual tendency to report local items systematically higher or lower, a form of subjective response style (Bond and Lang, 2019; Kapteyn et al., 2007; de New and Schurer, 2023). Because α_i enters only the local block, it absorbs whatever is common to a person’s local responses but left unexplained by observables, whether that reflects taste or reporting. Either way, allowing for α_i gives a more flexible and robust description of the data than the baseline that omits it.

C Model Fit

This appendix compares the observed distributions of the categorical survey responses with the distributions predicted by the model at the estimates of Section 5. For each response, the predicted share of a category is the average of the model-implied category probabilities across the individuals in the estimation sample. Table 12 reports the comparison for the three response blocks: the closeness of following each news type, the frequency of use of each local news provider, and the stated importance index. The table also compares the model-implied mean minutes of news per day by age group with the MCS anchor targets.

The predicted shares track the observed shares within about three percentage points in every category, across news types that share the same thresholds and across providers with very different usage patterns, and the model holds the age-group anchor means within about one percent.

D Data Validation Based on the ATUS

In this section, we compare our results with data from the American Time Use Survey (ATUS). While the ATUS does not collect detailed news consumption data, it provides comprehensive data on television watching for entertainment, allowing us to assess whether our data reflect broader media consumption patterns.

We compare three measures of TV consumption using identical control variables:

1. LNS: Categorical responses on local TV news consumption
2. Model Predictions: Daily minutes watching TV for news simulated from our structural model
3. ATUS: Daily minutes watching entertainment on TV

Table 13 reveals consistent demographic patterns across all measures, providing strong validation for our data and model. In particular, African Americans show

Table 12: Observed and Predicted Category Shares

<i>Panel A: closeness of following each news type</i>					
		Not at all	Not very	Somewhat	Very
Local	Observed	0.038	0.169	0.458	0.335
	Predicted	0.033	0.134	0.466	0.367
National	Observed	0.042	0.123	0.428	0.407
	Predicted	0.034	0.132	0.459	0.375
International	Observed	0.076	0.240	0.486	0.198
	Predicted	0.088	0.255	0.461	0.196
<i>Panel B: frequency of provider use</i>					
		Hardly ever	Sometimes	Often	
Newspaper	Observed	0.212	0.407	0.381	
	Predicted	0.223	0.401	0.375	
Television	Observed	0.125	0.330	0.545	
	Predicted	0.134	0.339	0.527	
Radio	Observed	0.243	0.443	0.314	
	Predicted	0.250	0.409	0.341	
Online media	Observed	0.175	0.375	0.450	
	Predicted	0.174	0.375	0.451	
Social media	Observed	0.259	0.383	0.359	
	Predicted	0.238	0.406	0.356	
<i>Panel C: stated importance index</i>					
		1 (low)	2	3	4 (high)
Index	Observed	0.009	0.197	0.732	0.062
	Predicted	0.009	0.197	0.731	0.062
<i>Panel D: age-group anchors (mean minutes of news per day)</i>					
		18–29	30–49	50–64	65+
Minutes	MCS target	45.00	67.15	76.00	83.00
	Model	44.94	66.69	76.23	83.99

Notes: Observed shares are the empirical category frequencies in the LNS estimation sample; provider frequencies are computed among respondents with the provider in their bundle. Predicted shares average the model-implied category probabilities at the estimates of Section 5 over the same individuals. The closeness and provider blocks share their thresholds across news types and providers, so their fit is not mechanical; the stated index margins are matched closely in part because its three cut-points are estimated freely. The age-group anchors in Panel D are targeted in the estimation through the penalty function, so that panel documents that the penalized fit holds the targeted means, rather than an out-of-sample comparison.

systematically higher TV consumption across all measures. They have 72% higher odds of frequent local TV news consumption (LNS), spend 4.52 additional minutes per day on local TV news (model), and watch 31.13 more minutes of entertainment TV daily (ATUS) compared to whites.

Educational patterns, age effects, and marital status show remarkable consistency across all measures, with college graduates, younger individuals, and married respondents consistently showing lower TV consumption for both news and entertainment. Some interesting dissimilarities emerge for Hispanics and by income group, which reflects higher information-seeking preferences rather than general TV viewing habits. Hispanics consume more TV news but less entertainment TV than whites, while higher income groups show increased news consumption but decreased entertainment consumption, suggesting these patterns reflect higher preferences for news content.

Table 13: Time Watching TV

	TV for news			TV for entertainment
	LNS	Model Local	Model All news	ATUS
	Categorical	Daily Minutes	Daily Minutes	Daily Minutes
Black	0.54*** (0.04)	4.52*** (0.11)	5.39*** (0.23)	31.13*** (4.18)
Hispanic	0.19*** (0.04)	2.70*** (0.10)	5.78*** (0.21)	-22.46*** (4.29)
Age	0.48*** (0.01)	1.46*** (0.03)	5.93*** (0.06)	2.92*** (0.08)
Male	-0.10*** (0.02)	-1.82*** (0.06)	2.87*** (0.12)	45.00*** (2.75)
Married	-0.08*** (0.02)	-1.54*** (0.06)	-3.08*** (0.13)	-23.89*** (2.95)
HS Graduate	-0.22*** (0.08)	-3.76*** (0.19)	-5.74*** (0.40)	-12.08** (5.11)
Some College	-0.38*** (0.07)	-5.12*** (0.19)	-8.01*** (0.38)	-35.92*** (5.05)
College Grad	-0.65*** (0.07)	-6.54*** (0.19)	-5.88*** (0.39)	-68.10*** (5.11)
Income Tercile 2	0.08*** (0.03)	0.44*** (0.08)	1.53*** (0.15)	-23.92*** (3.53)
Income Tercile 3	0.13*** (0.03)	0.82*** (0.08)	2.93*** (0.17)	-34.97*** (4.01)
Constant		8.17*** (0.62)	1.94 (1.27)	92.99*** (6.41)
City FE	Yes	Yes	Yes	Yes
Observations	27,352	27,352	27,352	17,879
R ²		0.26	0.30	0.14

Notes: The first column reports ordered logit coefficients for the frequency of local TV news in the LNS; the second and third columns report OLS coefficients for model-simulated daily minutes; the fourth column reports OLS coefficients for daily entertainment TV minutes in the ATUS. Standard errors in parentheses. *, **, and *** denote significance at the 10, 5, and 1 percent levels.